

Curriculum Vitae

JYOTSNA GIDUGU

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INTERESTS

I am a fourth-year PhD student at University of California San Diego, interested in the area of condensed matter theory. Specifically, I am interested in studying open quantum systems and in the fields of non-equilibrium many-body physics and quantum information.

RESEARCH SUMMARY

I am currently working on studying the **effects of dissipation in open quantum systems**. I have worked on projects that study the effects of dissipation on spin systems and fermionic systems. I had previously worked on **many-body localization in quantum systems** as a part of my Bachelor's dissertation and on **Josephson junction arrays** as a part of my Master's dissertation. I had also worked in the field of **experimental solid state physics** during the first three years of my undergraduate study.

EDUCATION

Currently pursuing a PhD in physics at **University of California San Diego (UCSD)** 2020 - Present

Completed a Master of Science degree at the **Indian Institute of Science (IISc), Bengaluru, India** 2019 - 2020

Completed a Bachelor of Science (Research) with a major in Physics and a minor in Chemistry at the **Indian Institute of Science (IISc), Bengaluru, India** 2015 - 2019

TEACHING EXPERIENCE

Served as a Teaching Assistant at UCSD for:
PHYS 2C (Fluids, Waves, Thermodynamics, and Optics), 1BL (Electricity and Magnetism Laboratory- online), 1AL (Mechanics Laboratory- online), 2CL (Electricity and Magnetism Laboratory), 100B (Electromagnetism II), 130A (Quantum Physics I), 1CL (Waves, Optics, and Modern Physics Laboratory), 110A (Mechanics I).

Served as a Lab teaching assistant coordinator (LTAC) at UCSD for:
Physics 1-series lab courses for Winter 2023, Spring 2023, Fall 2023 (current position).

PREPRINTS

Jyotsna Gidugu, Daniel P. Arovas. A study of dissipative models based on Dirac matrices. arXiv:2308.05245 [quant-ph]

PUBLICATIONS

Soumi Ghosh, **Jyotsna Gidugu**, and Subroto Mukerjee. Transport in the nonergodic extended phase of interacting quasiperiodic systems. Phys. Rev. B 102, 224203– Published 23 December 2020.

Pranava K. Sivakumar, Börge Göbel, Edouard Lesne, Anastasios Markou, **Jyotsna Gidugu**, James M. Taylor, Hakan Deniz, Jagannath Jena, Claudia Felser, Ingrid Mertig, and Stuart S. P. Parkin. Topological Hall Signatures of Two Chiral Spin Textures Hosted in a Single Tetragonal Inverse Heusler Thin Film. ACS Nano 2020, 14, 13463-13469.

Krishna Murthy J, **Jyotsna G**, N Nileena, Anil Kumar PS. Strain induced ferromagnetism and large magnetoresistance of epitaxial La_{1-x}Sr_xCoMnO₆ thin films. Journal of Applied Physics. 2017 Aug 14;122(6):065307.

CONFERENCES

Gave a talk and presented a poster at the **2023 Annual Meeting of the APS Far West Section held in the University of California, San Diego, California**. A study of dissipative models based on Dirac matrices, Jyotsna Gidugu and Daniel P. Arovas. October 6-7, 2023
Link: <https://sites.google.com/physics.ucsd.edu/apsfwsfallmeeting2023/home>
(Won the SPS Poster Award for a graduate student at the 2023 APS Far West Section Annual meeting at UCSD)

Presented a poster at the **Princeton Summer School on Condensed Matter Physics 2023: Fractionalization, criticality and unconventional quantum materials**. A study on dissipative models based on Γ -matrices, Jyotsna Gidugu and Daniel P. Arovas. July 10 - 14, 2023
Link: <https://pccm.princeton.edu/education/psscnp>

Attended and presented a poster at the **3rd Condensed Matter Summer School, "Dynamics and Quantum Information in Many-body Systems" at the University of Minnesota**. A study on dissipative models based on Γ -matrices, Jyotsna Gidugu and Daniel P. Arovas. June 12-21, 2023
Link: <https://cse.umn.edu/ftpi/cmss2023>

Presented a poster at a **conference at the Kavli Institute for Theoretical Physics on Topology, Symmetry and Interactions in Crystals: Emerging Concepts and Unifying Themes**. A study on dissipative models based on Γ -matrices, Jyotsna Gidugu and Daniel P. Arovas. Apr 3-6, 2023
Link: https://online.kitp.ucsb.edu/online/qcrystal-c23/gidugu/qcrystal-c23-poster_gidugu.pdf

RESEARCH EXPERIENCE

Effects of dissipation in open quantum systems
Have worked on studying the effects of dissipation on spin and fermionic systems. The project involved studying 2d dissipative models based on Dirac matrices. Under the supervision of **Prof. Daniel P Arovas, Department of Physics, University of California San Diego** as a part of my **PhD research**. 2021-Present

Granular superconductivity and high- T_c superconductors
Using Monte Carlo simulations of a 2D XY model to study magnetic and transport properties of high- T_c superconductors. Under the supervision of **Prof. H R Krishnamurthy and Prof. Subroto Mukerjee, Centre for Condensed Matter Theory, IISc** as a part of my **Master's dissertation**. 2019 August-Present

An investigation of thermalization and transport in a generalized

Aubry-Andre model.

Used computational methods such as exact diagonalization to study transport, level spacing statistics and entanglement in a system with deterministic disorder. Under the supervision of **Prof. Subroto Mukerjee, Centre for Condensed Matter Theory, IISc** as a part of my **Bachelor's dissertation**.

2018 Aug-
2019 July

Topological Hall effect in Heusler thin films of Mn_2RhSn

Studied the topological Hall effect in Heusler thin films of Mn_2RhSn to understand conditions that favor skyrmions and antiskyrmions. A manuscript on this work is in preparation to be submitted to Science Advances. Under the supervision of **Prof. Stuart Parkin** during my **summer internship at the Max Planck Institute of Microstructure Physics, Halle, Germany**.

2018 May-
2018 July

Structural and electronic properties of $h-BN$ and MoS_2

Studied the structural and electronic properties of some 2D materials like $h-BN$, MoS_2 and their hetero-structures using simulations and microscopy techniques. (URL for the report: <https://edu.authorcafe.com/academies/7020/structural-and-electronic-properties-of-h-bn-and-mos2>)

2017 May-
2017 July

Part of the **Indian Academy of Sciences summer fellowship (Science Academies' Summer Research Fellowship Programme-2017)**.

Under the guidance of **Prof. Ranjan Datta, at International Centre for Materials Science (ICMS), JNCASR, Bengaluru**.

Magnetic and transport properties of some double perovskites with antisite disorder

Studied the structural, magnetic, and magneto-transport properties of $La_{1.5}Sr_{0.5}CoMnO_6$ (LSCMO) thin films.

Under the guidance of **Prof. P S Anil Kumar at the magnetic and spintronic studies lab, IISc, Bengaluru** as a part of my summer project. **Resulted in a publication and a conference paper.**

2016 May-
2016 July

FELLOWSHIPS**Physics Excellence Award at UCSD**

Awarded in the first-year of duration at UCSD.

2020 Nov-
2021 July

Kishore Vaigyanik Protsahan Yojana [KVPY-SA]

Qualified in the KVPY exam in class 11.

Scholarship program funded by Department of Science and Technology, India.

2015 Aug-
Present

National Talent Search Examination [NTSE- 2011]

Qualified in the NTSE exam in class 8.

Awarded by NCERT, Govt. of India.

2011 - 2014

**RELEVANT
COURSES
STUDIED****Graduate level:**

Classical Mechanics, Quantum Mechanics, Mathematical Physics, Statistical mechanics, Electrodynamics, Condensed Matter Physics, Nonequilibrium statistical mechanics, Field Theory and Renormalization group, Dynamics: Deterministic, Stochastic, Statistical.

Undergraduate level:

Physics courses- Mechanics, Electromagnetism, Quantum Physics, Oscillations and waves, Thermal Physics and Physics of Materials

Math courses- Analysis, Linear Algebra, Probability and Statistics
Engineering courses- Algorithms and Programming, Electrical and Electronics Engineering, Materials Science

TECHNICAL SKILLS

Programming languages:
C, C++, Python

Mathematical packages:
ITensor (used to create tensor network algorithms), MATLAB, Mathematica,
WIEN2k (used to study crystal properties) *Computational skills:*
Strong knowledge of high-performance computing and MPI Parallel Programming.
Data analysis and visualization using Origin, Matplotlib.

Statistics:
Strong knowledge of Monte Carlo simulations, Langevin dynamics, studying statistics of eigenspectra of quantum systems, etc.

OTHER PROGRAMS

Research Science Initiative - 2014
Was among the few high school students selected to participate in the Research Science Initiative, Chennai- 2014.
Worked under the guidance of Dr. C V Krishnamurthy in the Department of Physics, IIT Madras during the summer of 2014 on a project on cloud chambers (particle detectors).